S-AI-CITY: EXPERIMENTAL EVALUATION OF HORMONAL URBAN ORCHESTRATION

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ABSTRACT

S-AI-CITY is a first-generation hormonal orchestration framework for urban intelligence in which heterogeneous urban observations are transformed by six Gland Agents into a shared regulatory state composed of Mobilin, Energexin, Environin, Resiliencin, Normin, and Servicin. Article I introduced the architecture, while Article II formalized its multi-layer urban graph, reaction-diffusion hormonal dynamics, 24 specialized agents, sparse constrained orchestration, hierarchical Zone → District → City organization, and symbolic Urban Engram memory. The present Article III defines the empirical evaluation stage of the trilogy. The study is organized around the SAI-UT+ CITY testbench and twelve parameterized urban scenario families spanning nominal operation, sector-specific stress, multi-domain disturbances, cascading failures, degraded communication, missing data, recovery, and recurrent events. The evaluation compares S-AI-CITY with a ladder of non-hormonal, rule-based, learned, model-based, random-budget, and offline-oracle references under matched quality-of-service and safety conditions. Decision quality, parsimony, operational cost, hormonal reactivity, cross-domain coordination, resilience, memory acceleration, explainability, constraint satisfaction, robustness, and scalability are evaluated jointly. The experimental design is explicitly falsifiable: each major architectural mechanism is associated with a null hypothesis and a corresponding ablation, and no benefit is attributed to a mechanism when its removal produces statistically indistinguishable behavior under matched conditions. Particular attention is given to avoiding circular validation by separating the scenario generator from the hormonal model and to preventing tuning asymmetry across competing methods. The resulting protocol is intended to determine not only whether S-AI-CITY can provide measurable operational advantages, but also which mechanisms produce those advantages, under which urban regimes they persist, and where they become negligible or disappear.

KEYWORDS

Sparse Artificial Intelligence; Smart Cities; Hormonal Orchestration; Multi-Stress Urban Systems; Experimental Evaluation; Sparse Agent Activation; Ablation Analysis; Quality of Service; Urban Resilience; Urban Engrams; Multi-Agent Systems.

1. INTRODUCTION

1.1. What Has Been Established in Articles I and II

S-AI-CITY was introduced as a first-generation Sparse Artificial Intelligence architecture for adaptive urban orchestration. Its central design principle is to avoid treating the city as a collection of permanently active analytical silos. Instead, heterogeneous urban observations are processed by six Gland Agents that generate artificial hormonal emissions corresponding to mobility, energy, environment and utilities, resilience, normative and security pressure, and public-service demand. These emissions are integrated by a Hormonal Engine into the six-dimensional regulatory state formed by Mobilin, Energexin, Environin, Resiliencin, Normin, and Servicin. A City-MetaAgent then uses this shared state, together with resource, safety, authority,

coverage, and temporal constraints, to activate only the specialized urban agents that are contextually relevant. The architecture is organized hierarchically across Zone, District, and City levels and maintains symbolic Urban Engrams for recurrent operational situations. Article II converted this architectural description into an explicit mathematical system. The urban substrate was represented as a dynamic multi-layer graph, glandular signals were expressed through bounded self-limiting emissions, and the Hormonal Engine was modeled as a reaction-diffusion system with leakage, cross-regulation, layer-dependent propagation, external regulation, delay, and stochastic perturbations. The same article formalized the complete registry of 24 specialized agents, their hormone-sensitivity vectors, contextual utilities, resource costs, authorization indicators, hysteresis, Time-to-Live, and Cooldown mechanisms, together with the constrained binary optimization problem used for parsimonious activation. It also separated numerical boundedness from continuous-time invariance and established conditional analytical results for deterministic forward invariance, frozen-input exponential stability, practical stability under persistent additive forcing, and bounded second moments under additive stochastic noise. These results characterize properties of the formal hormonal subsystem under stated assumptions; they do not by themselves establish superior urban performance. The role of Article III is therefore distinct. It does not redefine the architecture and does not reproduce the proofs of Article II. It asks whether the mechanisms already defined and formalized yield measurable empirical advantages when exposed to controlled, reproducible, and progressively more demanding urban conditions.

1.2. Objectives of Article III

The evaluation addresses ten complementary dimensions: decision quality, parsimony, hormonal reactivity, cross-domain coordination, resilience under degraded operation, memory acceleration during recurrent events, explainability and auditability, safety and constraint satisfaction, robustness to perturbation and distribution shift, and scalability. These dimensions are evaluated jointly because a reduction in active agents or computational cost is not meaningful if it is obtained by degrading service quality, violating critical constraints, or reducing resilience. The experiments are therefore designed around matched operating conditions whenever possible. Cost and parsimony are compared at comparable quality-of-service and safety levels; reactivity is examined together with overshoot, settling, and activation switching; memory benefits are evaluated together with decision quality; and scalability is assessed not only through runtime but also through preservation of the quality–cost trade-off as the urban substrate grows.

1.3. Central Falsifiable Thesis

The central empirical thesis of Article III is formulated as a falsifiable comparison rather than as a confirmatory assumption:
H_1: "at comparable QoS and safety,S-AI-CITY achieves a lower operational cost than a well-calibrated non-hormonal orchestration baseline."
The associated null hypothesis is
H_0: "at comparable QoS and safety,no statistically meaningful operational-cost advantage is observed."
The same logic is applied at mechanism level. Separate hypotheses are associated with hormonal cross-regulation, graph diffusion, hysteresis/TTL/Cooldown, symbolic memory, the Zone → District → City hierarchy, and the sparse activation objective. Each mechanism is paired with a corresponding ablation. A mechanism is not credited with an empirical benefit if its removal yields behavior that remains statistically indistinguishable from the complete system within the predefined confidence interval under matched QoS and safety conditions. This formulation also permits negative results to remain scientifically informative. A mechanism may improve performance only under multi-stress conditions, only above a given urban scale, or only when

communication, resource, or sensing conditions deteriorate. Conversely, a mechanism may provide no detectable advantage in nominal operation. Article III therefore seeks to identify the operating regimes in which the hormonal architecture contributes, rather than assuming that every mechanism should improve every metric in every scenario.

1.4. Contributions of Article III

Article III makes the experimental contribution of the S-AI-CITY trilogy through a reproducible SAI-UT+ CITY testbench, twelve parameterized multi-stress scenario families, an explicit separation between scenario generation and hormonal dynamics, a calibration protocol that applies comparable tuning effort to S-AI-CITY and its baselines, and a baseline ladder spanning always-on, threshold-based, rule-based, learned, model-based, random-budget, and offline-oracle references. The evaluation combines decision quality with a QoS–cost–safety analysis, mechanism-level ablations, parameter-sensitivity analysis, robustness and distribution-shift tests, repeated-run statistical analysis, and urban-scale variation. Together, these elements are intended to determine whether measurable benefits exist, to attribute them to specific architectural mechanisms when supported by the evidence, and to delimit the regimes in which the proposed hormonal orchestration remains useful, becomes neutral, or loses its advantage.

2. RELATED EXPERIMENTAL WORK

This section is deliberately narrower than the Related Work discussions of Articles I and II. Article I positioned S-AI-CITY within smart-city, multi-agent, bio-inspired, and explainable urban intelligence, whereas Article II reviewed the mathematical foundations required for the multi-layer graph, reaction-diffusion hormonal dynamics, stability analysis, stochastic integration, constrained sparse orchestration, and symbolic memory [1], [2]. Article III therefore restricts the literature review to experimental environments, comparator families, and evaluation practices that directly support the empirical protocol.

2.1. Scope and Non-Duplication with Articles I and II

The experimental problem differs from the architectural and mathematical questions addressed previously. The relevant issue is no longer whether a particular urban-control paradigm exists in the literature, but how competing orchestration strategies can be exposed to reproducible conditions, calibrated without systematic advantage, and compared through common outcome measures. Accordingly, the present section does not repeat the general smart-city state of the art or the theoretical foundations of S-AI-CITY. It focuses instead on simulation and co-simulation environments, representative rule-based, optimization-based, and learning-based comparators, and methodological principles for evaluating quality, resource use, robustness, coordination, and safety.

2.2. Urban Simulation, Co-Simulation, and Multi-Domain Testbeds

Urban-control research frequently relies on simulation because controlled repetition, stress injection, and direct comparison of alternative controllers are difficult to obtain in operational cities. In mobility, SUMO provides microscopic, multimodal traffic simulation and exposes programmable interfaces suitable for closed-loop control experiments [3]. For coupled infrastructures, HELICS illustrates a complementary co-simulation approach in which independently developed simulators exchange information at run time across domains that can include power, transportation, communications, and other energy-system components [4]. Such environments support higher-fidelity testing but also illustrate why an experimental framework

should keep the environment generator conceptually distinct from the controller being evaluated. Reproducible controller benchmarking is also well established in the energy domain. BOPTEST provides standardized simulation-based testing of building-control strategies through common emulators, scenarios, and performance indicators [5]. CityLearn similarly provides an environment for comparing rule-based, model-predictive, and reinforcement-learning control in grid-interactive communities, with multi-agent and resilience-oriented extensions in its second generation [6]. These precedents motivate the SAI-UT+ CITY design adopted in Article III: a controlled synthetic core, optionally anchored by one or more higher-fidelity domain simulators, with documented reduced-order models for the remaining urban domains.

2.3. Comparator Families for Urban Orchestration

The baseline ladder should represent distinct control principles rather than a single weak alternative. Rule-based and threshold controllers remain relevant because they provide transparent, low-complexity reference policies; model-predictive control provides a model-based optimization comparator; and learned controllers test whether adaptive selection can be obtained without an explicit hormonal state. BOPTEST and CityLearn both support comparisons across such controller families [5], [6]. In traffic control, PressLight showed how reinforcement learning can be coupled to a transportation-theoretic pressure formulation, while CoLight used graph-based communication to learn coordination among multiple intersections [7], [8]. These works justify including learned routing or reinforcement-learning comparators when they are functionally aligned with the S-AI-CITY selection problem. The objective is not to claim that any single baseline represents the complete state of the art, but to compare S-AI-CITY against several well-calibrated alternatives with comparable access to observations and comparable tuning effort.

2.4. Evaluation of Parsimony, Resilience, Coordination, and Safety

Existing benchmarking frameworks emphasize that controller comparisons require common scenarios and common performance indicators rather than isolated demonstrations [5], [6]. For S-AI-CITY, this requirement is extended to a joint evaluation of service quality, operational cost, sparse activation, cross-domain coordination, degraded-mode resilience, safety and authority constraints, memory-assisted recurrence, and decision traceability. In particular, a lower activation count or lower computational cost is not treated as an improvement when it is achieved through materially worse quality of service or increased violation of critical constraints. This motivates the QoS–Cost–Safety perspective adopted in the experimental protocol.

2.5. Positioning of the Present Evaluation

Article III therefore evaluates S-AI-CITY through a reproducible combination of

QoS + Cost + Safety + Multi-Stress + Ablation + Sensitivity + Robustness + Scalability.

The purpose of this combination is methodological rather than priority-seeking. The study is designed to determine whether the complete architecture occupies a favorable operating region relative to strong comparators, whether particular mechanisms account for any observed advantage, and whether those effects persist under parameter variation, degraded conditions, and increasing urban scale. Claims are therefore restricted to the evaluated scenarios and to the statistical evidence produced by the protocol.

3. EXPERIMENTAL QUESTIONS, HYPOTHESES, TESTBENCH, CALIBRATION, AND SCENARIOS

Article III treats the S-AI-CITY V1 architecture established in Articles I and II as a fixed experimental object. The six hormones, 24 specialized agents, Zone → District → City hierarchy, Hormonal Engine, City-MetaAgent, sparse-selection logic, and Urban Engram memory are not redesigned during evaluation [1], [2]. The experimental contract consists of falsifiable questions, a reproducible testbench, independent scenario generation, separated calibration and testing, and explicit annotation and performance oracles.

3.1. Evaluation Questions and Hypothesis Map

The evaluation is organized around seven principal questions. Each question is associated with a primary hypothesis H_i, a null hypothesis H_{0i}, a principal metric family, the scenario families in which the mechanism is expected to be most informative, and an ablation or comparison capable of challenging the hypothesis. The hypotheses concern experimental behavior within SAI-UT+ CITY; they are not reformulations of the conditional mathematical guarantees of Article II. **EQ1 — Decision Quality.** Does S-AI-CITY activate the specialized agents required by the controlled scenario semantics while avoiding unnecessary activations? The primary hypothesis H_1^{DQ} is that the complete architecture achieves higher agreement with the Ground-Truth Oracle than simple non-hormonal routing at comparable safety conditions. The null hypothesis H_0^{DQ} is that no statistically meaningful improvement is observed. Primary measures are precision, recall, F1, macro-F1 by agent family, and time-to-correct-activation. All scenario families contribute, with particular emphasis on multi-domain and cascading cases. The principal challenge is the No-Hormonal-Regulation ablation and the independent-threshold baseline. **EQ2 — Parsimony.** Can the architecture reduce unnecessary activation and operational cost without obtaining that reduction by accepting materially poorer service quality or safety? The primary hypothesis H_1^P is that, at matched QoS and safety, the sparse hormonal orchestration occupies a more favorable cost region than always-on and well-calibrated non-hormonal alternatives. The null hypothesis H_0^P states that the cost difference is not statistically meaningful under matched conditions. Primary measures are active-agent rate, computational and communication cost, quality per activation, and the QoS–Cost frontier subject to safety. The No-Parsimony-Penalty ablation and the always-on baseline provide the main counterfactual comparisons. **EQ3 — Cross-Domain Coordination.** Do the shared hormonal state and cross-regulation improve coordination when disturbances couple several urban sectors? The primary hypothesis H_1^{CD} is that the complete system produces fewer contradictory or poorly prioritized actions and better multi-domain service preservation than sector-isolated control. H_0^{CD} states that cross-domain outcomes remain statistically indistinguishable. Multi-stress families I, J, and K are primary, while F and G provide intermediate coupling cases. The No-Cross-Regulation ablation is the direct mechanism test; No-Hormonal-Diffusion provides a complementary test of spatial coordination. **EQ4 — Reactivity and Stability.** Do the six hormonal trajectories respond rapidly enough to changing urban pressure while avoiding excessive overshoot, persistent oscillation, repeated projection, or activation chattering? H_1^{RS} is that calibrated S-AI-CITY maintains an experimentally favorable reactivity–stability trade-off across nominal, peak, incident, and recovery phases. H_0^{RS} states that the observed dynamics do not differ meaningfully from simpler routing or that improved response is offset by excessive oscillatory behavior. Principal measurements include response latency, overshoot, settling time, oscillation count, projection frequency, pre-projection overshoot, switching frequency, and recovery time. The No-Hysteresis/TTL/Cooldown and No-Diffusion ablations are particularly informative. Article II provides the analytical quantities against which these observations can be interpreted, but empirical agreement is not treated as a

proof of those results [2]. **EQ5 — Memory Acceleration.** Do Urban Engrams provide useful acceleration when an event or a structurally similar event recurs? H_1^M is that memory-assisted retrieval and warm-start reduce time-to-correct-activation, solver effort, or recovery time without materially reducing decision quality or violating current constraints. H_0^M states that the memory-enabled and memory-disabled variants are statistically indistinguishable. Family L is the principal evaluation family, supplemented by recurrent instances drawn from B–K. The direct ablation disables Urban Engram retrieval while preserving the remainder of the architecture. Consistent with Article II, retrieval is treated as contextual assistance rather than automatic replay [2]. **EQ6 — Resilience and Degraded Mode.** Does the Zone → District → City organization preserve useful local service when higher-tier coordination, selected communication links, sensors, or resources become unavailable? H_1^R is that hierarchical S-AI-CITY retains a higher level of critical local continuity and recovers more rapidly than a flat architecture under controlled degradation. H_0^R states that the hierarchical and flat variants do not differ significantly. Primary measurements are service continuity, local-autonomy score, recovery time, escalation accuracy, critical coverage, and communication dependency. Scenario families F, G, H, K, and L are combined with explicit partition, missing-data, and resource-scarcity perturbations. The Flat-Architecture ablation provides the principal counterfactual. **EQ7 — Explainability, Safety, and Governance.** Can operational decisions be reconstructed from observable variables while hard safety, coverage, and authority constraints remain enforceable? H_1^{ESG} is that the architecture produces a complete decision trace from observations through glandular emissions, hormonal state, utilities, constraints, authorization gates, selected agents, actions, and outcomes for the evaluated decisions, with low rates of missing rationale and critical-constraint violation. H_0^{ESG} states that the proposed trace adds no measurable auditability or that constraint violations undermine the interpretation of sparse gains. Relevant metrics include trace completeness, authority violations, critical-coverage violations, infeasibility frequency, escalation frequency, and explanation overhead. All scenario families contribute, with H, I, K, and degraded-mode variants receiving particular attention. The seven questions are intentionally coupled: a parsimonious result is not interpreted favorably if decision quality or critical safety deteriorates materially. Matched-condition comparison therefore takes precedence over isolated optimization of any single metric.

3.2. SAI-UT+ CITY Testbench Architecture

SAI-UT+ CITY is defined as the controlled experimental environment in which the architecture is exercised. It extends the reproducibility objective already announced in Article I while preserving the exact S-AI-CITY V1 functional chain [1]:

Urban observations → Gland Agents → Hormonal Engine → $H(t)$ → City-MetaAgent → Specialized Urban Agents → Actions → Outcomes and Urban Engrams.

The initial nominal topology contains 12 urban zones grouped into three districts and one city tier. This 12-zone/3-district configuration is an experimental starting point rather than a claim about an optimal city partition. The nominal mapping uses four zones per district, but the assignment, graph density, and inter-district connectivity can be modified in the scalability experiments. Final values are frozen only after feasibility testing confirms that all baselines and ablations can be executed under comparable computational conditions. At the zone level, the urban substrate is represented by the same six functional layers formalized in Article II: mobility, energy/resources, environment/utilities, resilience/infrastructure, security/normative conditions, and public services [2]. The testbench therefore maintains a time-varying multi-layer graph

$$\mathcal{G}_C(t) = (\mathcal{V}, \mathscr{E}(t), \mathscr{L}, \mathcal{W}(t)),$$

where $\mathcal{V}$ denotes urban zones, $\mathscr{L}$ contains the six functional layers, and each layer may have a different adjacency pattern and edge weight structure. A mobility link need not imply an energy, water, cyber, or public-service link. This separation is important because the diffusion

mechanism tested in Article III is layer dependent rather than a generic propagation over a single graph. The experiment advances through synchronized decision cycles $t_n = n\Delta t$. Within each cycle, the environment first updates exogenous and endogenous urban variables, after which every evaluated controller receives an observation package derived from the same underlying urban state. For S-AI-CITY, these measurements are normalized into $X_i(t_n)$ with data-quality indicators $q_i^{data}(t_n)$; the six Gland Agents produce emissions; the Hormonal Engine updates the node-major hormonal state $H_n \in [0,1]^{6N}$; hierarchical summaries are formed when needed; Urban Engrams may be retrieved; the City-MetaAgent computes eligible agent utilities; and the constrained sparse-selection procedure returns the active subset. Outcomes are then applied to the environment before the next cycle. The registry of 24 specialized agents is not changed across nominal, robustness, and scalability experiments. Article III tests the architecture formalized in Article II rather than adding scenario-specific agents. Scalability therefore varies zones, districts, layer density, event concurrency, communication load, and memory size while retaining the same 24 agent types and their formal authorization envelopes [2]. Every run records a reproducibility manifest containing the scenario-family identifier, scenario parameters, random seed, graph realization, calibration profile, controller or ablation identifier, numerical step size, parameter set, and software configuration. The same scenario realization is replayed across competing methods whenever the method semantics permit it. This paired design reduces unnecessary environmental variance and helps attribute measured differences to the controller rather than to different underlying disturbances.

3.3. Two-Level Fidelity and Independence of the Scenario Generator

Where feasible, the simulation uses two fidelity levels. A higher-fidelity anchor domain can be supplied by a recognized simulator or public trace; mobility is a natural candidate because SUMO supports microscopic closed-loop traffic experiments [3]. Co-simulation frameworks such as HELICS illustrate how independently developed domain models can remain synchronized while retaining separate internal dynamics [4]. Remaining domains use documented reduced-order generators: demand/capacity processes for energy, reliability/restoration processes for infrastructure, environmental trajectories for heat and pollution, queue or service-loss models for utilities and public services, and independent attack or trust-degradation processes for cybersecurity. BOPTEST and CityLearn provide related precedents for controlled reproducible benchmarking [5], [6]. The key anti-circularity requirement is

$$G_{\text{urban}} \neq F_{\text{hormonal}}.$$

A congestion cascade is therefore not generated by C_H, and spatial stress is not propagated through the same hormonal diffusion coefficients $D_{k\ell}$ later evaluated by S-AI-CITY. Urban domains may still be causally coupled, but those couplings are specified by independent environmental rules or stochastic processes.

3.4. Data Sources

Level A is the core calibrated synthetic benchmark. It provides controlled stress intensity, onset, duration, failures, noise, missingness, communication quality, and ground-truth annotation, enabling counterfactual repetition for ablations and sensitivity analysis. **Level B** uses complementary public real-world traces where suitable data can be aligned with the testbench, including mobility, traffic, energy, environmental observations, or service demand. Level B is supplementary unless sufficiently broad multi-domain data are available; Level A remains the reproducible core and should not be described as real-city validation.

3.5. Parameter Calibration Protocol

Calibration is performed before the final evaluation and uses data that are disjoint from the reported test episodes. The objective is not to search for a single parameter vector that maximizes one headline metric, but to identify operational settings that preserve an acceptable compromise among decision quality, reactivity, stability indicators, resource use, and critical constraints. The calibration space follows the formal parameterization of Article II [2]. At minimum, it includes

$$\Theta_{\text{cal}} = \left\{\tau_k, \lambda_k, D_{k\ell}, C_H, \theta_j^{on}, \theta_j^{off}, \lambda_A, \tau_M\right\},$$

with delay parameters, noise amplitudes, data-quality fallback settings, numerical step size Δt, resource scalarization weights, TTL, and Cooldown included when they are active in the evaluated configuration. The ratio λ_k/τ_k is calibrated with response speed and dissipation in mind rather than treating λ_k and τ_k as unrelated knobs. Positive cross-regulation in C_H and diffusion coefficients $D_{k\ell}$ are calibrated jointly because stronger coupling may improve propagation or coordination while also changing the stability margin. Hysteresis width, TTL, and Cooldown are calibrated against switching frequency and response delay. The sparse penalty λ_A is calibrated against the QoS–Cost–Safety trade-off rather than against active-agent count alone. The memory threshold τ_M is calibrated to balance close contextual matching against broader recall. Article II provides several quantities that can serve as diagnostics rather than empirical claims: the dissipation margin μ, the persistent-input ratio $\bar{d}/\mu$, the reduced margin μ_{eff} under bounded state-dependent perturbation, the stochastic quantity q_Σ, projection frequency, and pre-projection overshoot [2]. Calibration can therefore distinguish parameter settings that remain comfortably inside the sufficient analytical envelope from settings that rely frequently on numerical projection. The forward-invariance conditions of Article II are treated as sufficient design conditions, not as necessary empirical boundaries. A limited number of parameter regimes that violate those sufficient inequalities may be retained deliberately for sensitivity experiments if the projected numerical implementation remains well defined. To limit tuning bias, the experimental data are separated into at least three roles: calibration, validation/model-selection, and final evaluation. A held-out distribution-shift regime is not exposed during tuning. Baselines receive comparable optimization effort, comparable access to observations, and documented search ranges. When a baseline has fewer parameters, fairness is sought through equivalent tuning opportunity rather than artificially increasing its complexity. The final manuscript reports the search domains, selected values, calibration objective, rejected unstable or infeasible configurations, and any controller-specific tuning budget. No final test episode is used to revise S-AI-CITY or baseline hyperparameters after its results are inspected.

3.6. Scenario Generator and Twelve Scenario Families

The twelve cases are **parameterized scenario families**, not single trajectories. A generic realization is described by

$$\vartheta^{sc} = (I, t_0, T, \Omega, p_f, \sigma_x, q_{\text{miss}}, q_{\text{comm}}, r_{\text{res}}),$$

representing stress intensity, onset, duration, spatial extent, failure probability, noise, missing-data level, communication quality, and available resources. Multiple seeds and parameter combinations are generated independently of $H(t)$.

Family	Principal stress	Dominant domains/hormones	Main experimental purpose
A	Nominal operation	Low multi-domain pressure	Reference overhead and switching
B	Morning mobility peak	Mobilin	Congestion response and propagation
C	Public transport	Mobilin + Servicin	Mobility/service coordination

Family	Principal stress disruption	Dominant domains/hormones	Main experimental purpose
D	Energy demand peak	Energexin	Energy response and sparse control
E	Heat and air quality	Environin	Environmental propagation and escalation
F	Water/utility incident	Environin + Resiliencin	Utility continuity and inter-layer response
G	Critical infrastructure failure	Resiliencin	Restoration, escalation, degraded mode
H	Cybersecurity incident	Normin	Authorization, containment, auditability
I	Mass public event	Mobilin + Servicin + Normin	Multi-sector coordination
J	Energy–mobility multi-stress	Energexin + Mobilin	Cross-domain regulation
K	Full urban cascade	Multiple hormones	Priority, arbitration, coverage, scarcity
L	Recovery and recurrence	Event-dependent	Memory acceleration and conditional reuse

Across the suite, the generator includes missing-data intervals, stochastic perturbations, resource scarcity, topology changes, communication partitions, cross-domain conflicts, recurrent events, and at least one held-out distribution-shift condition. Family K uses several cascade realizations rather than one hand-crafted sequence. Family L compares a first event with a later structurally similar recurrence at changed location, intensity, or operating context. Each scenario realization is frozen before controller execution and replayed across S-AI-CITY, baselines, and ablations.

3.7. Ground-Truth and Offline Oracles

Two distinct oracles are used because decision-quality annotation and performance normalization serve different experimental purposes. The first is the **Ground-Truth Oracle**$\mathcal{O}_{\mathrm{GT}}$,which maps the independently generated scenario state to annotation sets$\mathcal{A}_{\mathrm{req}}(t)$, $\mathcal{A}_{\mathrm{tol}}(t)$, $\mathcal{A}_{\mathrm{unnec}}(t)$,together with scenario-defined priorities, critical coverage classes, and prohibited or unauthorized actions. Required agents are those considered necessary by the benchmark semantics at that state; tolerated agents are defensible but non-essential; unnecessary agents have no scenario-grounded role. The oracle is specified before controller outputs are examined. It should preferably be derived from explicit scenario rules and independently reviewed mappings rather than from the hormonal scores themselves. Consequently,$\mathcal{O}_{\mathrm{GT}} \leftarrow /H(t)$,except where $H(t)$ is itself being assessed as an output. This restriction limits circular evaluation. The Ground-Truth Oracle is deliberately narrower than real municipal decision authority. Urban policy can be context dependent, legally constrained, and contested; therefore $\mathcal{O}_{\mathrm{GT}}$ is an experimental annotation instrument, not a claim that one mathematically defined activation set is the uniquely correct response for a real city. Ambiguous cases can be encoded through the tolerated set rather than forced into a binary correct/incorrect label. The second oracle is the **Offline Clairvoyant Oracle**$\mathcal{O}_{\mathrm{offline}}$,which is allowed access to the complete future trajectory of a scenario realization and computes, or approximates as closely as practical, the best attainable action or activation sequence under the same declared safety, authority, and resource constraints. Its purpose is to provide an upper normalization reference, not a deployable competitor. Because it uses future information unavailable online, it must never be described as a fair operational baseline. The separation between the two oracles is maintained

throughout the analysis:$\mathcal{O}_{\mathrm{GT}} \rightarrow$ decision-quality annotation, $\mathcal{O}_{\mathrm{offline}} \rightarrow$ performance upper reference.Together with a random budget-matched lower reference introduced later, these oracles permit performance to be interpreted relative to both scenario semantics and an approximate attainable envelope. Their definitions, constraint sets, and construction rules are fixed before the principal experiments so that they cannot be adjusted retrospectively to favor S-AI-CITY.

4. METRICS, SAFETY CRITERIA, AND BASELINES

Metrics are fixed before the principal experiments so that interpretation does not depend on selecting favorable measures after results are known. The evaluation follows the Article-I/II separation between decision relevance, parsimony, hormonal regulation, hard constraints, resilience, memory, and traceability [1], [2]. Lower cost or activation is favorable only when QoS and critical safety behavior remain acceptable.

4.1. Decision Quality Metrics

Decision quality is evaluated against the Ground-Truth Oracle $\mathcal{O}_{\mathrm{GT}}$ defined in Section 3.7. At cycle t, let $A(t)$ denote the set of agents selected by the evaluated controller, $\mathcal{A}_{\mathrm{req}}(t)$ the required-agent set, $\mathcal{A}_{\mathrm{tol}}(t)$ the tolerated-agent set, and $\mathcal{A}_{\mathrm{unnec}}(t)$ the unnecessary-agent set. Because tolerated activations are explicitly admissible, they are not counted as false positives. The operational precision and recall are therefore defined as

$$P(t) = \frac{|A(t) \cap (\mathcal{A}_{\mathrm{req}}(t) \cup \mathcal{A}_{\mathrm{tol}}(t))|}{|A(t)|}, \qquad R(t) = \frac{|A(t) \cap \mathcal{A}_{\mathrm{req}}(t)|}{|\mathcal{A}_{\mathrm{req}}(t)|},$$

with the conventional edge cases reported explicitly when either denominator is zero. The harmonic score is

$$F_1(t) = \frac{2P(t)R(t)}{P(t) + R(t)}.$$

Scores are first aggregated per run and then across repeated runs. Macro-F1 is computed over the six fixed agent families so that a large family or a frequently stressed domain does not dominate the global result. In addition, the unnecessary-activation rate

$$UAR = \frac{\sum_t |A(t) \cap \mathcal{A}_{\mathrm{unnec}}(t)|}{\sum_t |A(t)|}$$

is reported separately because it is directly relevant to parsimony. Time-to-correct-activation T_{corr} is measured from scenario onset to the first cycle at which all critical required agents are active and prohibited agents are absent. This temporal metric prevents a controller from appearing accurate solely because it eventually reaches the correct set after an operationally important delay.

4.2. Parsimony and Operational-Cost Metrics

The primary structural parsimony variable is the active-agent count

$$N_A(t) = |A(t)|,$$

with normalized activation rate

$$AR(t) = \frac{|A(t)|}{24}.$$

The mean, median, peak, and distribution of $AR(t)$ are reported because two controllers with the same average activation rate may differ substantially during crisis peaks. The 24-agent denominator remains fixed throughout the experiments, consistent with the Article II registry [2]. Operational cost is decomposed rather than represented initially by a single opaque scalar. For

each cycle, the benchmark records controller wall-clock time, agent-selection time, approximate computational operations or proxy FLOPs where meaningful, number of inter-tier or inter-agent messages, memory accesses, and an estimated computation/energy cost. If a composite cost is needed for Pareto analysis, normalized components are combined as

$$C_{\mathrm{op}}(t) = w_c C_{\mathrm{comp}}(t) + w_m C_{\mathrm{msg}}(t) + w_r C_{\mathrm{mem}}(t) + w_e C_{\mathrm{energy}}(t),$$

where $w_c, w_m, w_r, w_e \geq 0$ and $w_c + w_m + w_r + w_e = 1$. The component values and weights are always reported so that the composite score remains auditable. A complementary efficiency indicator is quality per activation,

$$QPA = \frac{\overline{Q}_{\mathrm{service}}}{\overline{N}_A + \varepsilon},$$

where $\varepsilon > 0$ only prevents division by zero. This indicator is interpreted jointly with the raw QoS and safety metrics: a high QPA obtained by under-serving a domain is not treated as a favorable result.

4.3. QoS Metrics by Urban Domain

Quality of service is measured separately for the principal urban domains before any aggregate score is constructed. Mobility uses travel time, congestion delay, queue length, or a normalized mobility-loss indicator. Energy uses unmet demand, load shedding, reserve deficit, or interruption duration. Environment and utilities use pollution/temperature exposure where applicable, water or utility interruption duration, and restoration performance. Resilience uses incident-response and recovery delay. Cybersecurity uses detection, containment, or trust-restoration latency under the scenario semantics. Public services use backlog, waiting time, or unresolved-demand fraction. For cross-domain comparison, each raw loss L_d for domain d can be mapped through a calibration-defined admissible range $[L_d^{best}, L_d^{worst}]$ to

$$Q_d = 1 - \mathrm{clip}\left(\frac{L_d - L_d^{best}}{L_d^{worst} - L_d^{best}}, 0, 1\right).$$

A global service score may then be formed as $Q_{\mathrm{service}} = \sum_d w_d Q_d$, with $w_d \geq 0$ and $\sum_d w_d = 1$. The domain scores and weights are reported alongside the aggregate. This prevents a strong gain in one sector from silently compensating for severe deterioration in another.

4.4. Hormonal Reactivity and Stability Metrics

Hormonal evaluation concerns measured trajectories rather than re-proving the analytical results of Article II. For each hormone k, peak latency is the interval between scenario onset and the first relevant local maximum; settling time is the time required to enter and remain within a predefined calibration-derived band around the post-event or recovery reference level. Overshoot is measured relative to that reference band, and oscillation count records repeated crossings beyond the tolerance region. When a calibration reference amplitude h_k^{ref} is available independently of the final test realization, amplitude error is

$$E_k^{amp} = |h_k^{peak} - h_k^{ref}|.$$

Propagation delay measures the time between a disturbance becoming observable in one zone and a statistically detectable hormonal response in another zone connected through the relevant layer. Cross-regulation fidelity is evaluated by checking whether observed signed co-variation and suppression/facilitation patterns are compatible with the configured nonzero entries of C_H; it is not interpreted as evidence that the city itself follows the hormonal coupling model. Temporal control is evaluated through activation-switching frequency and the distribution of agent holding times. Numerical behavior is characterized by projection frequency and pre-projection overshoot of the Euler-Maruyama proposal outside $[0,1]^{6N}$. Article II also permits a run-dependent estimate of the frozen dissipation margin,

$$\hat{\mu}(t) = \min_i \lambda_{\min}\left(\Lambda_H + \tilde{S}_i(t) - C_H^s\right),$$

together with an empirical practical-stability radius

$$r_{\text{emp}} = \text{quantile}_{0.95}\left(\| H(t) - H^{ref}(t) \|_2\right)$$

over a declared post-transient interval. These quantities are diagnostics. Agreement with the analytical trends of Article II may support consistency, but it does not convert empirical observation into a theorem [2].

4.5. Cross-Domain Coordination Metrics

On scenario states containing declared coupled priorities or conflicts, the study measures conflict-resolution accuracy, critical-priority preservation, contradictory simultaneous actions, arbitration/escalation frequency, required-agent coverage, and a multi-domain service score restricted to affected sectors. Hard incompatibility and coverage constraints are evaluated separately from softer coordination quality.

4.6. Memory Metrics

Memory evaluation separates retrieval quality from operational benefit. Metrics include engram retrieval success, relevance precision, warm-start success, time-to-correct-activation, recurrent-event acceleration, and solver-effort reduction where available. A simple acceleration factor is

$$G_{\text{mem}} = \frac{T_{\text{first}}}{T_{\text{recurrent}}}.$$

F1, QoS, cost, and safety are reported concurrently because faster recurrence is not beneficial if current-state constraints are degraded. Urban Engrams remain retrieval/warm-start mechanisms rather than correctness guarantees [2].

4.7. Resilience and Degraded-Mode Metrics

Resilience measures service continuity, area under degraded critical-service performance, recovery time, local-autonomy rate, escalation accuracy, and tolerance to district/city disconnection. Comparisons are made relative to the same controller's nominal behavior. These are empirical degraded-mode indicators and do not imply universal fault tolerance.

4.8. Safety, Authority, and Constraint-Satisfaction Metrics

Safety and authority are evaluated as first-class outcomes rather than absorbed into utility. The testbench records the number and rate of safety-constraint violations, critical-coverage violations, unauthorized activations, infeasible selection cycles, and fallback or human-escalation events. Authorization is evaluated against the hard gate $a_j(t) \le \rho_j(t)$ formalized in Article II, while critical coverage is evaluated against the active coverage classes required by the scenario [2]. For each violation type v,

$$R_v = \frac{N_v}{N_{\text{eligible cycles}}},$$

with both count and rate reported. Severity-weighted summaries may be added, but raw counts remain visible. A controller is not considered superior on parsimony or cost if its apparent gain is accompanied by materially worse critical-safety, authority, or coverage behavior. When no feasible automated action exists, correctly triggered fallback or escalation is treated differently from silent constraint violation.

4.9. Explainability and Audit Metrics

For each S-AI-CITY decision, the expected trace contains observations and data quality, glandular contribution, relevant hormonal state, utility/score, active constraints, authorization status, selected agents, override/arbitration events, and outcome. Trace completeness is

$$TC = \frac{N_{\text{present required fields}}}{N_{\text{required fields}}}.$$

Replay consistency, missing rationale, override provenance, and explanation-generation overhead are also measured. Architectural traceability is not equated with full semantic explainability of external learned components [1], [2].

4.10. Spatial Distribution Metric

For zone-level service scores Q_z, operational disparity is summarized by

$$D_{\text{sp}} = \overline{Q} - \min_z Q_z.$$

The worst-zone score is reported with the mean so that global improvement cannot silently hide severe local deterioration. This is an operational spatial-disparity measure, not a complete social-equity metric.

4.11. QoS–Cost–Safety Evaluation Protocol

The principal comparison is conducted under matched quality and safety whenever feasible. In the first protocol, a common QoS acceptance threshold $Q_{\min}$ and critical-safety envelope $\mathcal{S}_{\text{safe}}$ are imposed, and operational cost is compared only among runs satisfying

$$Q_{\text{service}} \geq Q_{\min}, \qquad x \in \mathcal{S}_{\text{safe}}.$$

In the second protocol, the resource or activation budget is swept to estimate the empirical Pareto set

$$\mathcal{P} = \text{Pareto}\left(Q_{\text{service}}, -C_{\text{op}}\right) \quad \text{subject to safety and authority constraints.}$$

A method is described as Pareto-dominant only on the tested points where it achieves no worse QoS and no greater cost, with at least one strict improvement, under the same safety requirements. The two views are complementary: matched-QoS comparison provides a directly interpretable cost difference, while the Pareto frontier reveals whether conclusions depend on a particular operating budget.

4.12. Baseline Ladder

All baselines receive the same underlying scenario realization, compatible observations, resource budget, and applicable hard constraints. **B1 — Always-On MAS:** all 24 authorized agents are active, providing a high-cost reference. **B2 — Independent Threshold Routing:** agents are activated directly from normalized observables without shared hormonal state, diffusion, cross-regulation, or memory. **B3 — Rules + Feature Flags:** a carefully calibrated deterministic controller with explicit priorities and safety rules. **B4 — Learned Router:** a compact learned gate maps observations to an active subset under the same hard constraints; learned traffic-control precedents illustrate the relevance of this comparator family [7], [8]. **B5 — Digital Twin + Optimization/MPC:** a model-based predictive reference when a sufficiently comparable implementation is feasible; BOPTEST and CityLearn illustrate benchmarking across rule-based, predictive, and learned controllers [5], [6]. **B6 — Random Budget-Matched Activation:** random authorized selection under the same average activation/resource budget, providing a lower normalization reference. **B7 — Offline Clairvoyant Oracle:** a future-aware upper

reference, never treated as an online deployable baseline. The S-AI-CITY-without-memory configuration is reserved for ablation rather than duplicated as a principal baseline.

5. MAIN EXPERIMENTAL RESULTS

The experiments reported in this section were executed only after the calibration protocol of Section 3.5 had been frozen. The final evaluation uses 20 Monte Carlo runs for each of the twelve scenario families, 56 orchestration cycles per run, 12 urban zones, three districts, and the fixed registry of 24 specialized agents defined in Article II [2]. Calibration and final-test seeds are disjoint. The S-AI-CITY threshold shift retained by the calibration search was -0.07; no parameter was modified after inspection of the final-test runs. All results in this section are therefore results of the controlled SAI-UT+ CITY simulation, not measurements from an operational city. The environment generator, semantic Ground-Truth Oracle, and hormonal dynamics remain distinct components, as specified in Section 3. The final test contains 1,920 method-by-family Monte Carlo runs and 107,520 principal controller cycles. The results are intentionally interpreted in both favorable and unfavorable directions. In particular, the experiments support a strong parsimony effect, a measurable recurrent-event memory benefit, and preservation of the hard safety envelope in the tested runs, but they do not support the stronger claim that the present S-AI-CITY configuration globally dominates all well-calibrated non-hormonal alternatives at matched QoS. This distinction is important because Article III was designed as a falsifiable evaluation rather than as a confirmatory demonstration.

5.1. Global Decision Quality

Table 1 reports the global decision-quality results. Confidence intervals are 95% intervals computed across the run-level aggregates. The offline oracle is included as an upper reference and should not be interpreted as an online competitor. Always-On obtains low F1 because activating almost the complete registry creates very low precision despite full coverage; its high QoS is analyzed separately in Section 5.2.

Method	Mean F1	95% CI	Mean QoS	Mean activation rate
Offline Oracle	0.9952	±0.0020	0.9104	0.0693
Learned Router	0.9906	±0.0023	0.9095	0.0802
Independent Threshold	0.9697	±0.0024	0.9102	0.0951
S-AI-CITY	0.9046	±0.0075	0.8990	0.0633
Rules + Feature Flags	0.8684	±0.0081	0.9104	0.1130
MPC Comparator	0.7323	±0.0095	0.9110	0.1311
Always-On	0.1390	±0.0128	0.9197	0.9988
Random Budget-Matched	0.0979	±0.0077	0.8688	0.1310

S-AI-CITY reaches a global mean F1 of 0.9046 ± 0.0075. This is *3.62* percentage points above Rules + Feature Flags and *17.23* points above the reduced-order MPC comparator, but *6.51* points below Independent Threshold and *8.60* points below the Learned Router. Its normalized F1 position between the Random and Offline-Oracle references is

$$\frac{F1_{\text{SAI}} - F1_{\text{Random}}}{F1_{\text{Oracle}} - F1_{\text{Random}}} = 0.8991.$$

The per-family results reveal that the global value does not arise from a single favorable scenario. S-AI-CITY reaches F1 $= 1.000$ in nominal Family A, *0.947* in Public Transport Disruption, *0.948* in Water/Utility Incident, *0.969* in Critical Infrastructure Failure, *0.947* in Mass Public Event, and *0.889* in the Full Urban Cascade. Its weakest values occur in Energy Demand Peak (*0.818*), Morning Mobility Peak (*0.846*), Energy–Mobility Multi-Stress (*0.859*), and Recovery/Recurrent Events when averaged over the complete family (*0.829*). Independent Threshold and the Learned

Router remain stronger on decision F1 in nearly every stressed family. These results therefore support a more limited statement than the central hypothesis initially contemplated. The hormonal architecture produces decision quality clearly above random, always-on precision, and the tested MPC formulation, and generally above or near the rule-based baseline, but the present calibration does not establish superior decision classification relative to the strongest threshold and learned selectors. This outcome is retained rather than removed because the purpose of the evaluation is to identify where the architecture does and does not add measurable value.

5.2. QoS–Cost–Safety Trade-Off and Parsimony

Parsimony is the clearest positive result of the principal experiment. S-AI-CITY activates on average *6.33*% of the 24-agent registry, corresponding to approximately *1.52* active agents per orchestration cycle. The activation rate is *33.47*% lower than Independent Threshold, *44.01*% lower than Rules + Feature Flags, *21.06*% lower than the Learned Router, and *51.74*% lower than the MPC comparator. Relative to Always-On, the reduction is *93.67*%, or approximately a *15.8* × reduction in active-agent fraction. This behavior is consistent with the parsimony objective defined architecturally in Article I and formalized as constrained sparse selection in Article II [1], [2]. The operational-cost index leads to a related but more nuanced result. S-AI-CITY obtains 2.758 ± 0.238 simulation cost units per cycle, compared with 2.909 ± 0.333 for Independent Threshold, 3.519 ± 0.365 for Rules + Feature Flags, 2.818 ± 0.285 for the Learned Router, 4.644 ± 0.369 for MPC, and 26.435 ± 0.114 for Always-On. The resulting cost reductions are approximately *5.2*%, *21.6*%, *2.1*%, *40.6*%, and *89.6*%, respectively. S-AI-CITY also obtains the highest quality-per-active-agent ratio among the evaluated online methods:$QPA_{\mathrm{SAI}} = 0.592$,versus *0.473* for the Learned Router, *0.399* for Independent Threshold, *0.336* for Rules + Feature Flags, and *0.290* for MPC. However, the QoS values prevent these savings from being interpreted as unconditional dominance. Mean S-AI-CITY QoS is 0.8990 ± 0.0025, whereas Independent Threshold, Rules, Learned, and MPC obtain *0.9102*, *0.9104*, *0.9095*, and *0.9110*, respectively. Thus S-AI-CITY saves between roughly *2*% and *41*% cost against these comparators, but does so with a global QoS decrease of approximately *1.05–1.20* percentage points. Under the strict protocol of Section 4.11, a cost advantage cannot be promoted to a matched-QoS superiority claim when the QoS difference remains visible. The empirical QoS–cost plane therefore contains an important negative result: S-AI-CITY is not globally Pareto-dominant in the current configuration. Independent Threshold has both lower cost and higher QoS than several alternatives and also retains higher QoS than S-AI-CITY, although its cost is modestly higher. The Learned Router similarly reaches higher QoS with only slightly higher cost. MPC occupies a higher-cost/higher-QoS region, while Always-On reaches the highest QoS but at an order-of-magnitude larger cost. The central thesis H_1, when interpreted strictly as lower operational cost at genuinely matched QoS and safety, is therefore **not established by the global aggregate**. What is supported is the narrower claim that S-AI-CITY achieves unusually sparse activation and a favorable quality-per-activation ratio while preserving the tested hard safety envelope. No safety, authority, or critical-coverage violation was recorded in the principal final-test cycles for S-AI-CITY or the constrained baselines. This zero count reflects the hard enforcement layer built into the experimental implementations; it should be interpreted as confirmation that the simulated constraint gates operated as specified, not as an independent proof of real-world safety.

5.3. Hormonal Trajectories and Reactivity

The six hormonal trajectories were recorded separately at every cycle and every Monte Carlo run. The Full Urban Cascade, Family K, provides the most informative joint stress test. Because its disturbances are spatially localized, zonal maxima are more representative of reaction amplitude

than city-wide means. The average run-wise zonal maxima reach their principal peaks at approximately cycle 33 for Mobilin (*0.351*), cycle 29 for Energexin (*0.388*), cycle 21 for Environin (*0.228*), cycle 34 for Resiliencin (*0.246*), cycle 36 for Normin (*0.174*), and cycle 34 for Servicin (*0.257*). The ordering follows the imposed independent cascade chronology rather than being hard-coded into the hormonal equations. The trajectories show three qualitative behaviors expected from the Article-II model [2]. First, localized stress produces a delayed rise rather than an instantaneous binary activation. Second, different hormones peak at different times according to the corresponding stress stream and time constants. Third, multi-stress families produce simultaneous nonzero hormonal components rather than forcing the city into a single categorical state. For example, the mean run-wise trajectory correlation in Family K is 0.598 ± 0.070 for Mobilin–Environin, 0.648 ± 0.069 for Energexin–Resiliencin, and 0.658 ± 0.119 for Normin–Servicin. In Family I, Mobilin–Servicin reaches 0.960 ± 0.015, while Normin–Servicin reaches 0.734 ± 0.066. These correlations are descriptive co-variation measures; they do not establish that the cross-regulation matrix is the causal source of the observed coupling. The conservative frozen-system dissipation lower bound computed from the calibrated leakage and symmetric cross-regulation terms is positive,$\hat{\mu}_{\text{lower}} = 0.7479$,which is consistent with a dissipative nominal parameterization. The empirical 95% spatial-spread proxy, computed from the distance between city mean and zonal maxima across recorded hormonal trajectories, is *0.1906*. These diagnostics are used only to relate the simulation to the conditional stability analysis of Article II; they do not re-prove Proposition 2. A less favorable numerical diagnostic is the frequency of projection. Depending on scenario family, approximately *77.7*%–*81.2*% of S-AI-CITY cycles contain at least one component of the unprojected Euler-Maruyama proposal outside [*0,1*]. The mean overshoot is nevertheless very small, approximately *0.0034–0.0043*. Inspection shows that most events correspond to shallow negative excursions of weakly excited hormones near the lower boundary rather than large positive saturation. The result suggests that the chosen stochastic amplitude and discretization are numerically well controlled in magnitude but rely on projection frequently. This is not evidence against the continuous-time stability result, because Article II explicitly separates projected numerical boundedness from invariance of the unprojected dynamics [2]. It does, however, identify a parameter region that deserves direct sensitivity analysis in Section 6.

5.4. Cross-Domain Coordination

Cross-domain coordination is evaluated on contested cycles, defined here as cycles in which at least two domain stresses exceed the declared multi-domain threshold. Two measures are reported: mean required-agent recall and the stricter **full-required-coverage** rate, which is one only when every oracle-required agent is simultaneously present. The latter is intentionally demanding and can therefore be low even when F1 remains high. In Mass Public Event (Family I), S-AI-CITY obtains contested-cycle mean recall *0.846*, F1 *0.914*, and full-required-coverage *18.6*%. Independent Threshold and Rules achieve full coverage on all contested cycles, while the Learned Router reaches *98.4*%. In Energy–Mobility Multi-Stress (Family J), S-AI-CITY improves to recall *0.859*, F1 *0.914*, and full coverage *53.8*%; the three strongest comparators again approach or reach full coverage. The Full Urban Cascade is more difficult: S-AI-CITY obtains contested-cycle recall *0.690*, F1 *0.810*, and full coverage *5.3*%, compared with *56.7*% for Independent Threshold, *60.0*% for Rules, *48.3*% for Learned, and *69.1*% for MPC. These results suggest that the shared hormonal representation remains capable of producing high partial agreement during multi-domain stress, but the present sparse selection setting can omit one or more lower-ranked required agents when the oracle simultaneously demands a broad coordinated response. The effect is particularly visible in Family K, where the same parsimony mechanism that reduces activation also limits all-required coverage. No contradictory action or hard coverage violation is recorded because the safety/coverage gate inserts mandatory critical elements before execution; nevertheless, the broader semantic coordination objective is not fully satisfied.

Accordingly, EQ3 is only partially supported by the main experiment. The observed multi-hormonal co-variation and high F1 indicate coherent shared-state response, but the strict full-coverage measure shows that cross-domain coordination is not uniformly superior to strong non-hormonal alternatives. The ablation of cross-regulation and hormonal diffusion in Section 6 is therefore necessary before any contribution can be attributed specifically to those mechanisms.

5.5. Memory Acceleration

Family L provides a paired first-occurrence/recurrent-event evaluation. In the memory-enabled S-AI-CITY configuration, mean time-to-correct activation decreases from 8.05 ± 1.57 cycles at the first occurrence to 5.90 ± 2.02 cycles at recurrence. The corresponding No-Memory configuration changes only from 7.75 ± 1.74 to 7.50 ± 1.79 cycles. Relative to the no-memory recurrent case, Urban Engram retrieval therefore reduces mean time-to-correct by *1.60* cycles, approximately *21.3*%, or a speed-up factor of about $1.27\times$. Decision quality also improves during the recurrent window. Memory-enabled S-AI-CITY reaches F1 *0.7750*, compared with *0.7089* for No-Memory, a gain of *6.61* percentage points. Recurrent-window QoS is *0.8953* with memory and *0.8893* without it. The gain is not free: the memory-enabled operational-cost index rises to *2.456* from *2.033*, and mean active agents rise from *1.014* to *1.156*, reflecting retrieval accesses and more anticipatory activation. A representative Family-L run stores five eligible engrams during the first event. At a recurrent episode, the maximum similarity reaches *0.997*, illustrating that the retrieval mechanism recognizes a structurally similar context. The retrieved memory remains advisory: present-state constraints and the current sparse selection still determine the final active set, consistent with the Article-II rule: retrieve ≠ replay.The evidence therefore supports a qualified form of EQ5: memory improves recurrent response time and F1 in this testbench, but at a measurable computation/activation cost. The claim should be framed as a speed–cost trade-off rather than as cost-free acceleration.

5.6. Degraded-Mode Resilience

Dedicated degradation experiments were executed on the Full Urban Cascade because it combines the highest coordination load with simultaneous infrastructure stresses. Nominal Family-K S-AI-CITY yields F1 *0.8894*, QoS *0.8421*, activation rate *0.2487*, and cost index *8.5205*. Five controlled degradation modes were then applied using disjoint random seeds. Loss of the City tier produces the largest degradation: F1 decreases by approximately *7.6*% to *0.822* and QoS by *5.0*% to *0.800*. A communication partition reduces F1 by *1.5*% and QoS by *1.2*%; District-level loss reduces them by *2.4*% and *1.7*%; and the missing-sensor condition reduces them by *1.2*% and *0.9*%. Under resource scarcity, F1 and QoS remain essentially unchanged relative to nominal K, while the cost index normalized by available resources increases by *17.6*%. The latter result reflects the intended interpretation of the cost index: performing a similar orchestration under a tighter resource envelope is more expensive in normalized terms. Across all five degraded experiments, no hard safety or critical-coverage violation is recorded. The result is compatible with the hierarchical design objective of Article I [1], but it should not be read as a universal graceful-degradation guarantee. The simulation implements explicit local safety fallback, so preservation of the hard envelope partly reflects the architecture's enforcement rules. The more informative empirical observation is that useful service continues with bounded QoS loss when communication or a district is impaired, whereas complete City-tier loss has a larger but still non-catastrophic effect in the tested cascade.

5.7. Explainability, Auditability, Authority, and Safety

Every final-test decision is stored with scenario identifier, seed, cycle, selected and required agent masks, decision metrics, cost, stress state, safety/authority status, and numerical projection diagnostics. For S-AI-CITY, separate hormonal traces are also retained. The instrumented audit run additionally records normalized observations, mean glandular emissions, mean and maximum hormonal state, agent utility scores, thresholds, contextual flags, and the final selected set. This provides an executable version of the architectural trace introduced in Articles I and II [1], [2]:

$$\text{observation} \rightarrow \text{glandular drive} \rightarrow H(t) \rightarrow U_j(t) \rightarrow \text{constraints} \rightarrow a_j(t) \rightarrow \text{outcome}.$$

A representative Full Urban Cascade trace occurs at cycle 29. The largest true stresses are Energexin-domain *0.696*, Mobilin-domain *0.565*, and Resiliencin-domain *0.506*. The corresponding maximum hormonal components are approximately *0.389* for Energexin, *0.288* for Mobilin, *0.114* for Resiliencin, *0.185* for Servicin, *0.142* for Environin, and *0.055* for Normin. Eleven agents are selected, including TrafficStateAssessmentAgent, GridBalanceAgent, UrbanHeatAgent, InfrastructureHealthAgent, EmergencyCoordinationAgent, FloodFireRiskAgent, UrbanCyberSecurityAgent, AccessAndTrustAgent, ServiceDemandAgent, and CrossDomainArbitrationAgent. The trace has precision *1.0*, recall *0.611*, and F1 *0.759*: every selected agent is admissible, but several oracle-required agents remain unselected. This example illustrates why auditability and decision completeness must be evaluated separately. No safety, authority, or hard critical-coverage violation is observed across the 107,520 principal final-test cycles. Again, this primarily confirms that the declared hard gates were correctly enforced in the simulator. It does not establish certification, legal compliance, or correctness of the semantic oracle. The important experimental contribution is reproducibility: the stored trace allows an external evaluator to reconstruct why an agent crossed its activation threshold, which constraints were active, and whether a fallback or projection event occurred.

5.8. Spatial Distribution

The spatial-distribution analysis tests whether favorable global averages hide systematically degraded zones. S-AI-CITY obtains mean zone QoS *0.8990*, mean worst-zone QoS *0.8523*, and

$$D_{\text{sp}} = 0.0467.$$

The corresponding disparities are *0.0458* for Independent Threshold, *0.0458* for Rules, *0.0459* for Learned, and *0.0457* for MPC. Always-On obtains *0.0456*. Differences are therefore small, but S-AI-CITY does not improve the spatial-disparity indicator in the present experiment and its worst-zone QoS is lower than that of the strongest baselines. The result does not indicate a large concentration of damage in one zone, because D_{sp} remains close to the comparator range. It nevertheless fails to support any claim that hormonal orchestration intrinsically improves spatial equity. The metric is operational rather than demographic or normative, so no broader fairness conclusion is drawn.

5.9. Summary of Main Experimental Findings

Table 2 summarizes the experimental claims at this stage. “Supported” is intentionally reserved for findings directly visible in the final-test evidence; “partial” indicates that some relevant metrics improve while another part of the stated hypothesis remains unresolved or unfavorable.

Claim	Main evidence	Strong comparator/result	Status
Decision quality	S-AI F1 *0.9046* ± *0.0075*	Threshold *0.9697*, Learned *0.9906*	Partial
Parsimony	Activation rate *0.0633*	33–52% lower than	Supported

Claim	Main evidence	Strong comparator/result	Status
		Threshold/Rules/MPC	
Matched QoS–cost superiority	Cost *2.758*, QoS *0.8990*	Strong baselines retain QoS ≈ *0.910*	Not supported globally
Hormonal reactivity	Distinct stress-dependent peaks; positive $\hat{\mu}_{lower}$	Frequent but shallow projection events	Partial
Cross-domain coordination	Multi-stress F1 remains high	Strict full coverage weaker, especially Family K	Partial
Memory acceleration	Recurrent TTC *5.90* vs *7.50* cycles; +6.61 pp F1	Higher retrieval/activation cost	Supported with cost caveat
Degraded-mode continuity	QoS loss mostly $<$ *2%*, except City loss ≈ *5%*	Zero hard violations in tested runs	Partial
Auditability and hard constraints	Reconstructible traces; zero hard violations	Enforcement partly structural by design	Supported within testbench
Spatial-distribution advantage	$D_{\mathrm{sp}} = 0.0467$	Strong baselines *0.0457–0.0459*	Not supported

The main empirical conclusion is therefore more restrained than a simple statement that S-AI-CITY outperforms conventional orchestration. In this controlled configuration, the framework exhibits a strong and measurable activation-parsimony advantage and obtains the highest quality-per-active-agent ratio, while Urban Engrams improve recurrent-event response and the hierarchical safety envelope remains operational under several degradation modes. At the same time, the strongest threshold and learned comparators achieve higher global decision F1 and QoS, strict multi-domain full coverage remains weaker for S-AI-CITY in the full cascade, projection is frequent although shallow, and no spatial-distribution advantage is observed. These mixed findings motivate the mechanism-level ablations, parameter-sensitivity analysis, and robustness study of Section 6 rather than supporting a universal superiority claim.

6. ABLATIONS, SENSITIVITY, ROBUSTNESS, STATISTICS AND SCALABILITY

The analyses in this section use random seeds disjoint from the calibration and final-test seeds of Section 5. Their purpose is not to maximize the headline performance of S-AI-CITY, but to determine whether the principal mechanisms introduced architecturally in Article I and formalized in Article II remain experimentally distinguishable when individual components are removed, parameters are perturbed, operating conditions shift, or the urban substrate grows [1], [2]. The results should therefore be interpreted as mechanism-level evidence within the SAI-UT+ CITY prototype rather than as universal properties of hormonal orchestration. Two implementation limits are made explicit. First, the current experimental selector realizes parsimony through an effective activation bias rather than solving the complete Article-II constrained binary program at every cycle; A5 is consequently an implementation-level proxy for removing the sparse penalty. Second, TTL and Cooldown are not separately instantiated in the current prototype, so A6 isolates the implemented hysteresis term. The flat-architecture ablation A7 likewise suppresses hierarchical/local criticality through city-wide mean aggregation and should be interpreted as a controlled proxy rather than a complete alternative deployment stack.

6.1. Mechanism-Level Ablation Matrix

Seven ablations were evaluated against the same Full-SAI configuration over the twelve scenario families, with 20 repeated runs per family. A1 replaces the hormonal state by direct observable-driven activation. A2 sets the off-diagonal hormonal cross-regulation to zero,$C_H^{\text{off}} = 0$.A3 removes hormonal propagation by setting$D_{k\ell} = 0$.
A4 disables Urban Engram retrieval and memory-guided warm-start. A5 suppresses the effective sparsity bias of the implementation by lowering the activation barrier, serving as the prototype counterpart of removing the λ_A-type parsimony term. A6 removes the implemented hysteresis contribution to the on/off threshold. A7 replaces the max-preserving spatial summary by a single city-wide mean before agent selection. Table 3 reports the principal aggregate outcomes.

Configuration	F1	QoS	Activation rate	Cost index	Switching frequency
Full S-AI-CITY	0.9054	0.8992	0.0631	2.752	0.1161
A1 — No Hormonal Regulation	0.4394	0.9115	0.1759	5.763	0.8173
A2 — No Cross-Regulation	0.9055	0.8986	0.0618	2.716	0.1133
A3 — No Hormonal Diffusion	0.9006	0.8996	0.0645	2.788	0.1191
A4 — No Memory	0.9028	0.8990	0.0631	2.744	0.1165
A5 — No Parsimony Penalty proxy	0.2830	0.9121	0.3047	9.055	0.0907
A6 — No Hysteresis	0.9196	0.8977	0.0575	2.605	0.1264
A7 — Flat Architecture proxy	0.9268	0.8981	0.0610	2.703	0.1102

A1 produces the largest degradation attributable to removal of the shared hormonal state: F1 decreases by approximately *46.6* percentage points relative to Full S-AI-CITY, while activation rises from *6.31*% to *17.59*% and switching frequency increases by a factor of about seven. QoS nevertheless rises to *0.9115*, because the direct controller activates substantially more agents. The result therefore supports a regulatory/parsimony role for the hormonal state more strongly than a QoS-superiority claim. The paired F1 difference is large ($d_z = 2.95$, Holm-adjusted $p < 10^{-39}$). A2 is deliberately informative because it yields essentially no global F1 change: *0.9055* without cross-regulation versus *0.9054* for the full system. The paired difference is not significant after Holm correction ($p = 0.600$). Under the present scenario distribution, the global decision-quality benefit therefore cannot be attributed to C_H alone. Cross-regulation may still affect particular multi-domain trajectories, but the experiment does not support treating it as a dominant global mechanism. A3 produces a small F1 reduction of about *0.48* percentage point and a slight increase in activation and cost. The paired effect is statistically detectable (Holm-adjusted $p = 2.5 \times 10^{-7}$) but modest in magnitude ($d_z = 0.33$). This suggests that graph diffusion contributes measurably in the current benchmark, although its practical effect is much smaller than that of the complete hormonal representation. A4 changes global F1 by only *0.26* percentage point and is not significant after correction ($p = 0.150$). This does not contradict the targeted recurrent-event result of Section 5: memory is active mainly in Family L, so its benefit is diluted when averaged across all twelve families. The more specific evidence remains the reduction of recurrent time-to-correct from *7.50* to *5.90* cycles and the recurrent-window F1 gain reported in Section 5. A5 illustrates the cost of removing the sparse preference in the current implementation. Activation

increases from *6.31*% to *30.47*%, the cost index rises from *2.752* to *9.055*, and QoS improves from *0.8992* to *0.9121*. F1 falls sharply because the enlarged set contains many oracle-unnecessary agents and therefore loses precision. This ablation supports the importance of an activation penalty for parsimony, but the numerical result should not be interpreted as an exact experiment with the full Article-II $\lambda_A = 0$ binary optimizer because the prototype uses threshold-equivalent sparse gating [2]. A6 exposes a genuine trade-off. Removing hysteresis increases F1 from *0.9054* to *0.9196* and lowers activation and cost, but switching frequency rises from *0.1161* to *0.1264*, approximately *8.9*%. The temporal smoother therefore reduces chattering at the price of some delayed deactivation or retained activations. Since TTL and Cooldown are not separately instantiated here, no claim is made about their individual contribution. A7 is also a negative or neutral finding. The flat-aggregation proxy reaches higher F1 (*0.9268*) than the full configuration while producing similar QoS and activation. The normal-operation experiment therefore does not establish a global classification advantage from the hierarchy itself. Article I motivates Zone → District → City organization principally through locality and degraded connectivity [1], and the resilience experiments of Section 5 remain more relevant to that claim than this mean-aggregation proxy. A full distributed-versus-flat implementation comparison remains necessary before attributing a general performance gain to hierarchy.

6.2. Parameter Sensitivity Analysis

A one-factor-at-a-time sensitivity study was executed on seven representative families B, D, E, G, I, K, L, using four disjoint seeds per setting. The tested neighborhood covers

$$\left\{\lambda_k/\tau_k,\ C_H,\ D_{k\ell},\ \theta_j^{on/off},\ \lambda_A^{\text{eff}},\ \tau_M,\ \sigma,\ \delta,\ q^{data}\right\},$$

where λ_A^{eff} denotes the prototype's effective sparsity/threshold bias. Cross-regulation and diffusion are comparatively insensitive in the explored range. Scaling C_H from *0.5* to *1.5* of its nominal value leaves mean F1 within *0.8883–0.8885*; scaling diffusion over the same range yields *0.8853–0.8883*. The response ratio is somewhat more influential: scaling leakage relative to timescale from *0.8* to *1.2* moves F1 from *0.8700* to *0.8901*, while activation decreases from *8.63*% to *7.33*%. The effective activation threshold shows the expected quality–parsimony compromise. Threshold shifts $-0.10, -0.07, -0.04$ yield F1 values *0.8664,0.8883,0.8695*, activation rates *9.56*%, *7.80*%, *6.70*%, and QoS values *0.8995,0.8930,0.8864*, respectively. The calibrated -0.07 setting lies near the best F1 region rather than at either parsimony extreme. Hysteresis width is more sensitive: widths *0.020,0.045,0.070* produce F1 *0.9076,0.8883,0.8483*. This reinforces the A6 result that temporal smoothing should be calibrated jointly with decision latency and switching, rather than maximized independently. Observation delay is the strongest tested sensitivity. Zero-, one-, and three-cycle delays reduce F1 from *0.8883* to *0.8570* and *0.7905*, respectively, with corresponding QoS decline. Hormonal noise has little effect on F1 over amplitudes *0.0045–0.018*, but it strongly changes the numerical projection diagnostic: projection frequency rises from *49.3*% to *90.4*%, and mean pre-projection overshoot from approximately *0.0007* to *0.0110*. Thus task-level robustness can coexist with numerically less comfortable trajectories. Observation noise up to *0.070* does not reduce aggregate F1 in this experiment, partly because additional uncertain activations increase coverage; this should not be generalized beyond the tested generator. Varying the fallback value *0.10–0.30* and the memory threshold $\tau_M = 0.72$–*0.88* produces little global change, indicating that these parameters are weakly identified by the selected aggregate benchmark even though they can matter in targeted missing-data or recurrence episodes. Overall, performance is not confined to an extremely narrow neighborhood for C_H, diffusion, or hormonal-noise amplitude, but it is materially more sensitive to delay, hysteresis width, and sparse-selection threshold. Those dimensions should therefore receive priority in any later deployment-specific calibration.

6.3. Robustness and Distribution Shift

Robustness tests use Family K as the principal stress anchor and compare it with sensor, communication, topology, resource, and distribution-shift perturbations. Nominal K in this disjoint robustness set gives F1 *0.8955* and QoS *0.8446*. With *45*% high-missingness injection during the stressed interval, F1 remains *0.8883* and QoS decreases to *0.8305*. A *30*% random edge removal yields F1 *0.8876* and QoS *0.8457*, suggesting limited sensitivity to this level of topology damage in the tested graph family. High sensor noise reduces F1 to *0.8711*, while severe communication degradation reduces it to *0.8702* and QoS to *0.8172*. The communication case is therefore more damaging to service quality than missingness or topology loss. The resource-shock condition leaves F1 (*0.8901*) and QoS (*0.8442*) nearly unchanged, but the normalized cost index increases from *8.34* to *13.39*, reflecting the tighter resource envelope rather than additional activations. The strongest degradation appears under the unseen combined distribution shift, where stress magnitudes, temporal ordering, missingness, communication quality, and resources are jointly changed without recalibration. F1 falls to *0.8441* and QoS to *0.7912*, approximately *5.7*% and *6.3*% below nominal K. Family L, which was excluded from the calibration families, obtains F1 *0.8314* and QoS *0.9037*; its different operating semantics prevent direct comparison with K, but it provides an additional held-out distribution. The robustness evidence is therefore conditional. The architecture tolerates moderate isolated missingness and topology perturbation reasonably well in the testbench, whereas communication degradation, delay, and compound unseen shifts remain more consequential. The results favor describing S-AI-CITY as exhibiting bounded robustness within the simulated neighborhood rather than claiming distribution-independent resilience.

6.4. Repeated Runs and Statistical Protocol

The principal Section-5 comparisons use 240 matched run-level observations per method: twelve families multiplied by 20 controlled seeds. S-AI-CITY and each comparator receive the same family/seed realization. For each primary metric, the paired difference distribution is checked before selecting the test. The principal F1 differences are non-normal under the Shapiro–Wilk diagnostic, so the Wilcoxon signed-rank test is used; families of pairwise tests are corrected by the Holm procedure. Effect size is reported through paired Cohen's d_z and rank-biserial correlation so that statistical significance is not confused with practical magnitude. For F1, S-AI-CITY is significantly lower than Independent Threshold (-6.51 percentage points, $d_z = -1.27$, Holm $p = 2.3 \times 10^{-37}$) and the Learned Router (-8.60 points, $d_z = -1.42$, $p = 2.3 \times 10^{-37}$). It is significantly higher than Rules ($+3.62$ points, $d_z = 0.69$, $p = 7.6 \times 10^{-23}$), MPC ($+17.23$ points, $d_z = 1.92$, $p = 2.4 \times 10^{-40}$), and Always-On precision ($+76.56$ points, $d_z = 6.22$, $p = 2.0 \times 10^{-40}$). These tests reinforce rather than replace the descriptive trade-offs of Section 5. The ablation statistics are equally informative. A2 No-Cross-Regulation is not distinguishable from Full S-AI-CITY on global F1 after correction ($p = 0.600$); A4 No-Memory is also not significant globally ($p = 0.150$). A3 No-Diffusion has a small but detectable effect ($d_z = 0.33$), whereas A1 and A5 have very large effects. Removing hysteresis and using the flat proxy significantly improve F1 in the aggregate, which is precisely why those results are retained as counter-evidence rather than interpreted selectively. Primary hypotheses, calibration settings, and final-test data were frozen before these Section-6 analyses, while the ablation/sensitivity seeds were generated in separate ranges.

6.5. Scalability Study

Scalability keeps the 24 agent types fixed and varies the substrate and workload. The principal size sweep uses 12, 24, 48, and 96 zones with 3, 6, 12, and 24 districts, while the number of

simultaneous stresses and memory entries also increases to create a progressively harder workload. The measured core cycle time rises from approximately *0.216* ms at 12 zones to *0.309* ms at 96 zones. Hormonal diffusion time rises from *0.089* to *0.135* ms, while selection time remains comparatively stable, *0.086* to *0.098* ms, consistent with the fixed 24-agent registry described in Article II [2]. The communication-load proxy grows more strongly, from approximately *185* to *2297* messages/link operations per cycle. The heavier workload also changes behavior: activation rate rises from *19.7*% at 12 zones to *31.6*% at 96 zones, F1 decreases from *0.972* to *0.913*, and QoS from *0.920* to *0.878*. Thus the experiment does not show an increasing **relative** parsimony advantage with scale: the fraction of agents avoided relative to Always-On decreases from about *80.3*% to *68.4*% as simultaneous stress increases. Parsimony remains substantial, but the present benchmark does not establish that its relative value necessarily grows with city size. One-factor scalability tests clarify where cost accumulates. At 48 zones, increasing the extra-edge probability from *0.01* to *0.12* approximately doubles the communication proxy from *724* to *1496*, while measured diffusion time changes little because the current Python implementation uses dense vectorized adjacency matrices. These timings therefore should not be interpreted as a proof of sparse-graph asymptotic complexity. Increasing simultaneous events from one to six raises activation from *18.0*% to *31.5*%, while F1 decreases from *0.987* to *0.920* and QoS from *0.934* to *0.877*. Increasing memory from 0 to 5000 engrams raises measured retrieval time from approximately *0.0001* ms to *0.259* ms and core cycle time from *0.188* to *0.468* ms, indicating that unindexed exhaustive memory search can become a non-negligible scaling bottleneck. The scalability results therefore support a narrower conclusion than simple linear scalability. With a fixed 24-agent registry, selection cost remains relatively stable, whereas communication, graph processing, event concurrency, and unindexed memory retrieval become increasingly important. Future large-scale implementations should consequently exploit sparse graph operators, indexed or approximate memory retrieval, and distributed hierarchy rather than relying on the dense prototype used for this controlled study.

7. DISCUSSION, LIMITATIONS, THREATS TO VALIDITY AND DEPLOYMENT

7.1. Theoretical Interpretation of Experimental Results

The results are broadly compatible with several qualitative expectations of Article II but do not constitute proofs of its analytical propositions [2]. The projected hormonal field remains bounded, the conservative dissipation diagnostic is positive, and the six components show differentiated responses to independently generated stresses. Yet the ablations are more informative than correlation alone: removing the complete hormonal representation strongly degrades F1 and increases activation and switching, whereas suppressing cross-regulation alone changes global F1 little and removing diffusion has only a modest effect. The evidence therefore supports the usefulness of the shared hormonal state more clearly than the necessity of every individual term. S-AI-CITY also achieves the lowest activation rate among the principal online controllers and the highest quality-per-active-agent ratio, consistent with the sparse orchestration objective [2]. However, Threshold and Learned retain higher global F1 and QoS. Urban Engrams improve recurrent response but have little global effect because recurrence occupies only part of the testbench. The empirical picture is consequently mechanism- and regime-dependent.

7.2. Comparison with the State of the Art

Comparison is restricted to experimentally supported dimensions. Against the tested rule-based, threshold, learned-routing, predictive-control, always-on, random, and offline-oracle references, S-AI-CITY is particularly competitive in activation parsimony, quality per active agent, traceability, and recurrent-event acceleration, but it does not lead the strongest baselines in global

F1 or QoS. This is consistent with benchmarking practice in environments such as BOPTEST and CityLearn, where controller quality is assessed through shared scenarios and multiple indicators rather than one headline score [5], [6]. Strong learned traffic controllers likewise illustrate that domain-specific learning can be highly effective for narrowly specified objectives [7], [8]. S-AI-CITY should therefore be positioned as a multi-domain orchestration study rather than as a claim of general superiority. The evidence does not justify "first," "unique," or "only existing framework" claims, nor preference over every optimized domain-specific controller.

7.3. What S-AI-CITY Adds Beyond S-AI-IoT

S-AI-CITY retains first-generation S-AI hormonal regulation, selective activation, specialized agents, and symbolic memory, but moves the orchestration problem from constrained device networks to a macroscopic network of interacting urban infrastructures. Mobility, energy, environment/utilities, resilience, security/normative pressure, and public services coexist on distinct graph layers [1], [2]. This introduces heterogeneous infrastructure topology, Zone → District → City aggregation, inter-infrastructure conflicts, localized stress propagation, urban-service arbitration, and authority constraints. The cascade and degraded-mode experiments suggest that these interactions are meaningful, although the flat proxy does not prove that hierarchy improves global F1. The added value is therefore the formulation and testing of urban-scale multi-infrastructure orchestration rather than a claim that every urban extension outperforms its antecedent.

7.4. Practical Deployment Path

A prudent deployment path is progressive: **(1)** shadow mode with no operational influence; **(2)** advisory recommendations reviewed by operators; **(3)** a narrow class of reversible automated actions with explicit rollback; **(4)** district-level integration after local validation; **(5)** inter-district coordination; and **(6)** broader automation only under explicit governance, auditable logs, human override, cybersecurity review, and domain-specific authorization or certification where applicable.

7.5. Internal Validity

The Ground-Truth Oracle is an experimental annotation mechanism, not an objective model of municipal correctness; alternative admissible mappings could change precision and F1. Calibration and scenario selection may favor regimes naturally represented by the six hormonal dimensions despite separated calibration and test seeds. Baseline strength also depends on implementation depth: the Learned Router and reduced-order MPC are controlled comparators, not exhaustive realizations of their research fields, and exact tuning symmetry is difficult across models with different inductive biases. Prototype-specific threats also matter. A5 approximates removal of the sparse penalty through the implemented activation bias rather than executing the complete Article-II binary optimization with $\lambda_A = 0$. A6 isolates implemented hysteresis but not separately instantiated TTL/Cooldown. A7 is a flat-aggregation proxy rather than a full alternative architecture. Simulator circularity is reduced because environmental coupling and stress propagation are independent from C_H and hormonal diffusion, but the benchmark and controller still share the same six urban semantic domains.

7.6. External and Construct Validity

The principal external limitation is the simulation-to-reality gap. The results come from calibrated synthetic and reduced-order processes and omit much of the non-stationarity,

institutional constraints, heterogeneous sensing, human adaptation, maintenance, and adversarial behavior of an operational city. Twelve zones and three districts are structurally limited; even the 96-zone scalability study uses synthetic graphs and dense numerical operators. QoS, activation rate, cost proxies, latency, safety violations, and spatial disparity also do not fully represent accessibility, trust, distributive justice, externalities, or citizen well-being. An audit trace demonstrates architectural traceability but does not guarantee understandable, socially acceptable, or legally sufficient explanation. Likewise, the spatial metric detects local deterioration but is not a complete equity model.

8. General Synthesis and Conclusion of the Trilogy

8.1. Contributions Across the Three Articles

The trilogy progresses from architecture to formalization and empirical evaluation. **Article I** defines the six urban hormones—Mobilin, Energexin, Environin, Resiliencin, Normin, and Servicin—together with Gland Agents, Hormonal Engine, City-MetaAgent, Zone → District → City hierarchy, specialized-agent families, Urban Engrams, and governance constraints [1]. **Article II** formalizes the multi-layer graph, fixes the 24-agent registry, develops glandular emissions, reaction-diffusion dynamics, cross-regulation, conditional stability and perturbation bounds, projected stochastic discretization, constrained sparse orchestration, authorization/coverage rules, temporal activation mechanisms, and Engram retrieval [2]. **Article III** adds falsifiable experiments, the SAI-UT+ CITY testbench, twelve multi-stress families, strong baselines, QoS–Cost–Safety evaluation, ablations, sensitivity, robustness, statistics, scalability, and explicit validity limits.

8.2. Central Result to Emphasize

The central result is not universal cost superiority. S-AI-CITY **can occupy a favorable parsimony region of the QoS–Cost–Safety trade-off in specific regimes**: it activates about *6.3*% of the 24-agent registry globally, obtains the highest quality-per-active-agent ratio among evaluated online methods, preserves the implemented hard safety envelope, and benefits from Urban Engrams during recurrence. At the same time, Threshold and Learned obtain higher global F1 and slightly higher QoS; strict multi-domain full coverage is weaker in the full cascade; cross-regulation has little global F1 effect in the present ablation; the flat proxy does not underperform hierarchy on global classification; and delay and compound distribution shift remain significant weaknesses. The evidence therefore supports a conditional, not universal, advantage.

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